

Recipe Generator - Example Output

Step 1: Diet & Health Score Filter

At startup, the code displays a list of available dietary preferences (gluten-free, halal, kosher, nut-free, vegan). The user can select one to filter the recipes, or press Enter to skip. The user is then asked to set a minimum health score between 0 and 100. In this example the user entered 60, meaning only recipes with a health score of 60 or above will be returned.

```
/usr/local/bin/python3 /Users/xenabezuidenhout/Documents/PRA2031/Recipe-Generator-PRA2031/src/generator.py
○ xenabezuidenhout@Xenas-MacBook-Air Recipe-Generator-PRA2031 % /usr/local/bin/python3 /Users/xenabezuidenhout/Documents/PRA2031/Recipe-Generator-PRA2031/src/generator.py
Available diet options:
1. gluten_free
2. halal
3. kosher
4. nut_free
5. vegan
Select a diet option (or press Enter to skip):
Enter minimum health score (0-100, or press Enter to skip): 60
```

Step 2: User Ingredient Input

The code presents a “Quick Ingredient Input” screen where the user types all available ingredients in one line, separated by commas. The system automatically categorises them (proteins, carbs, produce, dairy, pantry, produce).

QUICK INGREDIENT INPUT

Enter all your ingredients separated by commas.
They will be automatically categorized.

Your ingredients:

> chicken, broccoli, cheese, flour, onion, tamatoes, carrots, rice, potatoes

Step 3: Top 10 matching recipes

The code searches the used database for the recipes and ranks the best matches by how many of the user's ingredients each recipe uses (the match score). The top 10 results are displayed. The user can then select a recipe to view its full details (in this example, the user selected 5-Grilled Chicken Burgers).

```
Top 10 recipes:
1.Savory Roasted Root Vegetables (match score: 3)
2.Vegan Split Pea Soup (match score: 3)
3.Leftover Roast Chicken Soup (match score: 3)
4.Best of All: Slow Cooker Beef Stew (match score: 3)
5.Grilled Chicken Burgers (match score: 3)
6.Ukrainian Salat Vinaigrette (Beet Salad) (match score: 3)
7.Turkey Carcass Soup (match score: 3)
8.Yam and Turnip Stew with Mini-Biscuits (match score: 3)
9.Hearty Slow Cooker Beef Stew (match score: 3)
10.Savory Apple Cider Beef Stew (match score: 3)
Select a recipe to view details (1-10): 5
```

Step 4: Full recipe detailed view

The selected recipe is displayed in full. This includes the recipe name, its health score, and all diet tags it satisfies. The code then splits the ingredient list into two sections: ingredients the user already has and ingredients that are missing. Finally, the full step-by-step cooking directions are shown.

```
Recipe: Grilled Chicken Burgers
Health Score: 84.0/10 (healthy)
Diet Tags: gluten_free, nut_free, halal, kosher

Ingredients you have (by category):
PRODUCE:
- 1 medium onion, chopped
- 2 medium carrots, chopped
PROTEINS:
- 2 pounds ground chicken

Missing ingredients (by category):
CARBS:
- ½ cup fresh bread crumbs
PANTRY:
- cooking spray
- 1 medium tomato, seeded and chopped
- 1 tablespoon seafood seasoning (such as Old Bay®)
- kosher salt and ground black pepper to taste
PRODUCE:
- 2 teaspoons minced garlic
- 1 medium red bell pepper, chopped
- 1 cup fresh sliced mushrooms
PROTEINS:
- 1 large egg

Directions:
1. Lightly spray a skillet with cooking spray and heat over medium heat. Add onion and sauté until translucent, about 5 minutes. Add garlic and cook until fragrant, about 30 seconds. Add carrots, bell pepper, tomato, and mushrooms; cook and stir until tender, 5 to 7 minutes. Remove from the heat and let cool completely, about 20 minutes
2. Preheat an outdoor grill for medium heat and lightly oil the grate
3. Mix ground chicken and cooled vegetables together in a large bowl. Add bread crumbs, egg, seafood seasoning, salt, and pepper; mix until well combined. Form mixture into eight patties
4. Cook on the preheated grill until no longer pink in the center and the juices run clear, 5 to 6 minutes per side. An instant-read thermometer inserted into the center should read at least 165 degrees F (74 degrees C).
```